Linear Algebra

Lecture Notes

Row Space, Column Space and Linear Systems

Topics covered:

- Row Space and Column Space
- Linear Systems
- Consistency Theorem for Linear Systems
- Rank–Nullity Theorem
- The Column Space

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Contents

1. Subspaces Associated with a Matrix

Every matrix $\mathbf{A} \in \mathbb{R}^{m\times n}$ naturally gives rise to two fundamental subspaces: the *row space* and the *column space*. Understanding these subspaces is central to the theory of linear systems.

1.1 Row Space

Definition 1.1 (Row Space)

Let $\mathbf{A}$ be an $m \times n$ matrix with rows $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_m \in \mathbb{R}^{1\times n}$. The **row space** of $\mathbf{A}$, denoted $\mathrm{Row}(\mathbf{A})$, is the subspace of $\mathbb{R}^n$ spanned by the rows of $\mathbf{A}$:

$$\mathrm{Row}(\mathbf{A}) = \mathrm{Span}\{\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_m\}.$$

Proposition 1.1 Row Operations Preserve the Row Space

Elementary row operations on a matrix $\mathbf{A}$ do *not* change the row space:

$$\mathrm{Row}(\mathbf{A}) = \mathrm{Row}(\mathbf{R}),$$

where $\mathbf{R}$ is any matrix obtained from $\mathbf{A}$ by elementary row operations (e.g. the reduced row echelon form).

Remark 1.1

This proposition means we can find a basis for $\mathrm{Row}(\mathbf{A})$ by row reducing $\mathbf{A}$ to (reduced) row echelon form $\mathbf{R}$ and taking the *non-zero rows* of $\mathbf{R}$. These non-zero rows form a basis for $\mathrm{Row}(\mathbf{A})$.

Example 1.1 Finding a Basis for the Row Space

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

Row reduce $\mathbf{A}$:

$$\mathbf{A} \xrightarrow{R_2 \leftarrow R_2 - 4R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 7R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix}.$$

The non-zero rows of the row echelon form are $\mathbf{r}_1 = (1, 2, 3)$ and $\mathbf{r}_2 = (0, -3, -6)$.

Therefore

$$\mathrm{Row}(\mathbf{A}) = \mathrm{Span}\big\{(1,2,3),\ (0,-3,-6)\big\}, \quad \dim\big(\mathrm{Row}(\mathbf{A})\big) = 2.$$

1.2 Column Space

Definition 1.2 (Column Space)

Let $\mathbf{A}$ be an $m \times n$ matrix with columns $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n \in \mathbb{R}^m$. The **column space** of $\mathbf{A}$, denoted $\mathrm{Col}(\mathbf{A})$, is the subspace of $\mathbb{R}^m$ spanned by the columns of $\mathbf{A}$:

$$\mathrm{Col}(\mathbf{A}) = \mathrm{Span}\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}.$$

Remark 1.2

$\mathrm{Col}(\mathbf{A}) = \mathrm{Row}(\mathbf{A}^T)$. Thus a basis for the column space of $\mathbf{A}$ can be obtained by finding a basis for the row space of $\mathbf{A}^T$. Equivalently, the *pivot columns* of $\mathbf{A}$ (identified by row reduction) form a basis for $\mathrm{Col}(\mathbf{A})$, using the **original** columns of $\mathbf{A}$, not those of the row-reduced form.

Example 1.2 Finding a Basis for the Column Space

Using the matrix from the previous example,

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}.$$

From the row reduction, the pivot positions are in columns 1 and 2. Therefore a basis for $\mathrm{Col}(\mathbf{A})$ is formed by columns 1 and 2 of the *original* matrix:

$$\mathrm{Col}(\mathbf{A}) = \mathrm{Span}\left\{ \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} \right\}, \quad \dim\big(\mathrm{Col}(\mathbf{A})\big) = 2.$$

2. Linear Systems

2.1 Formulation

A **system of linear equations** (or *linear system*) in n unknowns $x_1, x_2, \ldots, x_n$ is a collection of m equations of the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

where $a_{ij}, b_i \in \mathbb{R}$.

2.2 Matrix–Vector Form

The system above can be written compactly as

$$\mathbf{A}\mathbf{x} = \mathbf{b},$$

where $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{m\times n}$ is the **coefficient matrix**, $\mathbf{x} = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$, and $\mathbf{b} = (b_1, \ldots, b_m)^T \in \mathbb{R}^m$.

Definition 2.1 (Augmented Matrix)

The **augmented matrix** of the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is the $m \times (n+1)$ matrix

$$[\,\mathbf{A} \mid \mathbf{b}\,].$$

Row operations on the augmented matrix correspond exactly to legal operations on the linear system and preserve the solution set.

2.3 Solution Types

A linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$ has exactly one of the following:

- **No solution** (inconsistent),
- **Exactly one solution** (consistent, unique),
- **Infinitely many solutions** (consistent, non-unique).

Example 2.1 Solving a Linear System by Row Reduction

Solve the system

$$\begin{cases} x_1 + 2x_2 + x_3 = 4 \\ 2x_1 + 3x_2 - x_3 = 1 \\ 3x_1 + 5x_2 \qquad = 5 \end{cases}$$

Augmented matrix and row reduction:

$$\begin{pmatrix} [ccc|c]1 & 2 & 1 & 4 \\ 2 & 3 & -1 & 1 \\ 3 & 5 & 0 & 5 \end{pmatrix} \to \begin{pmatrix} [ccc|c]1 & 2 & 1 & 4 \\ 0 & -1 & -3 & -7 \\ 0 & -1 & -3 & -7 \end{pmatrix} \to \begin{pmatrix} [ccc|c]1 & 2 & 1 & 4 \\ 0 & 1 & 3 & 7 \\ 0 & 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} [ccc|c]1 & 0 & -5 & - \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

Free variable: $x_3 = t \in \mathbb{R}$. The general solution is

$$\mathbf{x} = \begin{pmatrix} -10 \\ 7 \\ 0 \end{pmatrix} + t \begin{pmatrix} 5 \\ -3 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

3. Consistency Theorem for Linear Systems

The following theorem gives a clean characterisation of when a linear system has a solution, phrased entirely in terms of the column space of the coefficient matrix.

Theorem 3.1 Consistency Theorem

The linear system $\mathbf{Ax} = \mathbf{b}$ is **consistent** (i.e. has at least one solution) if and only if

$$\mathbf{b} \in \mathrm{Col}(\mathbf{A}).$$

Equivalently, the system is consistent if and only if

$$\mathrm{rank}(\mathbf{A}) = \mathrm{rank}([\,\mathbf{A} \mid \mathbf{b}\,]).$$

Remark 3.1

The second form of the Consistency Theorem is practically useful:

- Row reduce the augmented matrix $[\mathbf{A} \mid \mathbf{b}]$.
- If a row of the form $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$ appears, the system is *inconsistent.*
- Otherwise the system is *consistent.*

Theorem 3.2 Uniqueness of Solutions

Suppose $\mathbf{Ax} = \mathbf{b}$ is consistent. Then the solution is **unique** if and only if $\mathrm{Null}(\mathbf{A}) = \{\mathbf{0}\}$, i.e. the only solution to $\mathbf{Ax} = \mathbf{0}$ is the trivial solution. Equivalently, uniqueness holds if and only if $\mathrm{rank}(\mathbf{A}) = n$ (no free variables).

Example 3.1 Checking Consistency

Is the system

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{A} = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

consistent? Row reduce the augmented matrix:

$$\begin{pmatrix} \text{[cc|c]}1 & 3 & 2 \\ 2 & 6 & 5 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} \text{[cc|c]}1 & 3 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

The last row represents $0 = 1$, a contradiction. Since $\mathrm{rank}(\mathbf{A}) = 1 < 2 = \mathrm{rank}([\mathbf{A} \mid \mathbf{b}])$, the system is **inconsistent**. Equivalently, $\mathbf{b} \notin \mathrm{Col}(\mathbf{A})$.

Corollary 3.1 Consistent System for all b

The system $\mathbf{Ax} = \mathbf{b}$ is consistent for *every* $\mathbf{b} \in \mathbb{R}^m$ if and only if

$$\mathrm{Col}(\mathbf{A}) = \mathbb{R}^m \iff \mathrm{rank}(\mathbf{A}) = m.$$

4. Null Space and the Rank–Nullity Theorem

4.1 Null Space

Definition 4.1 (Null Space)

The **null space** (or *kernel*) of an $m \times n$ matrix $\mathbf{A}$ is the set of all solutions to the homogeneous system:

$$\mathrm{Null}(\mathbf{A}) = \{\, \mathbf{x} \in \mathbb{R}^n : \mathbf{Ax} = \mathbf{0} \,\}.$$

Proposition 4.1 Null Space is a Subspace

$\mathrm{Null}(\mathbf{A})$ is a subspace of $\mathbb{R}^n$.

Proof. (i) $\mathbf{0} \in \mathrm{Null}(\mathbf{A})$ since $\mathbf{A0} = \mathbf{0}$.
(ii) If $\mathbf{u}, \mathbf{v} \in \mathrm{Null}(\mathbf{A})$, then $\mathbf{A}(\mathbf{u} + \mathbf{v}) = \mathbf{Au} + \mathbf{Av} = \mathbf{0} + \mathbf{0} = \mathbf{0}$.
(iii) If $\mathbf{u} \in \mathrm{Null}(\mathbf{A})$ and $c \in \mathbb{R}$, then $\mathbf{A}(c\mathbf{u}) = c\mathbf{Au} = c\mathbf{0} = \mathbf{0}$. □

Definition 4.2 (Rank and Nullity)

Let $\mathbf{A}$ be an $m \times n$ matrix.

- The **rank** of $\mathbf{A}$, written $\operatorname{rank}(\mathbf{A})$, is the dimension of $\operatorname{Col}(\mathbf{A})$ (equivalently, of $\operatorname{Row}(\mathbf{A})$):
$$\operatorname{rank}(\mathbf{A}) = \dim\big(\operatorname{Col}(\mathbf{A})\big) = \dim\big(\operatorname{Row}(\mathbf{A})\big).$$
- The **nullity** of $\mathbf{A}$, written $\operatorname{nullity}(\mathbf{A})$, is the dimension of $\operatorname{Null}(\mathbf{A})$:
$$\operatorname{nullity}(\mathbf{A}) = \dim\big(\operatorname{Null}(\mathbf{A})\big).$$

Remark 4.1

$\operatorname{rank}(\mathbf{A})$ equals the number of *pivot columns* (leading entries) in the row echelon form of $\mathbf{A}$. The number of *free variables* equals the nullity.

4.2 The Rank–Nullity Theorem

Theorem 4.1 Rank–Nullity Theorem (Dimension Theorem)

Let $\mathbf{A}$ be an $m \times n$ matrix. Then
$$\boxed{\operatorname{rank}(\mathbf{A}) + \operatorname{nullity}(\mathbf{A}) = n.}$$

In words: the number of pivot variables plus the number of free variables equals the total number of unknowns.

Sketch of Proof. Let $r = \operatorname{rank}(\mathbf{A})$. Row reduce $\mathbf{A}$ to row echelon form. There are r pivot columns and $n - r$ free columns. Each free variable contributes one basis vector (a *special solution*) to $\operatorname{Null}(\mathbf{A})$, so $\operatorname{nullity}(\mathbf{A}) = n - r$. Therefore $r + (n - r) = n$. □ □

Example 4.1 Applying the Rank–Nullity Theorem

Let
$$\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

This matrix is already in row echelon form. There are $r = 2$ pivot columns (columns 1 and 3) and $n = 4$ unknowns. By the Rank–Nullity Theorem:
$$\operatorname{rank}(\mathbf{A}) = 2, \qquad \operatorname{nullity}(\mathbf{A}) = 4 - 2 = 2.$$

The free variables are $x_2 = s$ and $x_4 = t$. Back-substituting gives the general homogeneous solution:

$$\mathbf{x} = s\begin{pmatrix}-2\\1\\0\\0\end{pmatrix} + t\begin{pmatrix}1\\0\\-3\\1\end{pmatrix}, \qquad s,t \in \mathbb{R}.$$

A basis for $\mathrm{Null}(\mathbf{A})$ is $\left\{\begin{pmatrix}-2\\1\\0\\0\end{pmatrix}, \begin{pmatrix}1\\0\\-3\\1\end{pmatrix}\right\}$.

Corollary 4.1 Invertibility Criterion

An $n \times n$ square matrix $\mathbf{A}$ is **invertible** if and only if

$$\mathrm{rank}(\mathbf{A}) = n \iff \mathrm{nullity}(\mathbf{A}) = 0.$$

5. The Column Space in Depth

5.1 Characterisation via Linear Combinations

A vector $\mathbf{b} \in \mathbb{R}^m$ belongs to $\mathrm{Col}(\mathbf{A})$ if and only if there exist scalars $c_1, \ldots, c_n$ such that

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \cdots + c_n\mathbf{a}_n = \mathbf{b},$$

i.e. if and only if the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ is consistent. This re-states the Consistency Theorem in a geometric language.

5.2 Basis for the Column Space

Theorem 5.1 Pivot Column Theorem

Let $\mathbf{R}$ be the row echelon form of $\mathbf{A}$. The **pivot columns of $\mathbf{A}$** (the original columns corresponding to pivot positions in $\mathbf{R}$) form a basis for $\mathrm{Col}(\mathbf{A})$.

Remark 5.1

Important: Use the pivot columns from $\mathbf{A}$, not from $\mathbf{R}$. The column space of $\mathbf{R}$ is generally *different* from $\mathrm{Col}(\mathbf{A})$, because column operations change the column space (unlike row operations, which do not change the row space).

Example 5.1 Complete Column Space Analysis

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & -1 & 3 \\ 4 & 2 & -2 & 6 \\ 1 & 0 & 1 & 2 \end{pmatrix}.$$

Step 1: Row reduce.

$$\mathbf{A} \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 2 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 2 & 1 & -1 & 3 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Step 2: Identify pivots. Pivot columns are 1 and 2.

Step 3: Basis for $\mathrm{Col}(\mathbf{A})$.

$$\text{Basis} = \left\{ \mathbf{a}_1 = \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}, \quad \mathbf{a}_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right\}.$$

Step 4: Summary.

$$\mathrm{rank}(\mathbf{A}) = 2, \quad \mathrm{nullity}(\mathbf{A}) = 4 - 2 = 2, \quad \dim\big(\mathrm{Col}(\mathbf{A})\big) = 2.$$

5.3 Relationship Between the Four Fundamental Subspaces

For an $m \times n$ matrix $\mathbf{A}$, the four *fundamental subspaces* are:

Subspace	**Notation**	**Lives in**
Row space	$\mathrm{Row}(\mathbf{A})$	$\mathbb{R}^n$
Null space	$\mathrm{Null}(\mathbf{A})$	$\mathbb{R}^n$
Column space	$\mathrm{Col}(\mathbf{A})$	$\mathbb{R}^m$
Left null space	$\mathrm{Null}(\mathbf{A}^T)$	$\mathbb{R}^m$

Theorem 5.2 Orthogonality of Fundamental Subspaces

$$\mathrm{Row}(\mathbf{A}) \perp \mathrm{Null}(\mathbf{A}) \quad \text{in } \mathbb{R}^n,$$

$$\mathrm{Col}(\mathbf{A}) \perp \mathrm{Null}(\mathbf{A}^T) \quad \text{in } \mathbb{R}^m.$$

Moreover,

$$\dim\big(\mathrm{Row}(\mathbf{A})\big) + \dim\big(\mathrm{Null}(\mathbf{A})\big) = n,$$

$$\dim(\mathrm{Col}(\mathbf{A})) + \dim(\mathrm{Null}(\mathbf{A}^T)) = m.$$

Remark 5.2 Visual Summary

The Rank–Nullity Theorem and the Consistency Theorem connect all five objects: $\mathbf{A}$, $\mathbf{b}$, $\mathrm{Col}(\mathbf{A})$, $\mathrm{Null}(\mathbf{A})$, and $\mathrm{rank}(\mathbf{A})$. Together they give a complete picture of when and how many solutions a linear system possesses.

6. Summary and Key Results

Key Results at a Glance

1. **Row space basis:** Non-zero rows of the row echelon form of $\mathbf{A}$.
2. **Column space basis:** Pivot columns of the *original* $\mathbf{A}$.
3. **Consistency Theorem:** $\mathbf{Ax} = \mathbf{b}$ is consistent $\iff$ $\mathbf{b} \in \mathrm{Col}(\mathbf{A})$ $\iff$ $\mathrm{rank}(\mathbf{A}) = \mathrm{rank}([\mathbf{A} \mid \mathbf{b}])$.
4. **Rank–Nullity Theorem:** $\mathrm{rank}(\mathbf{A}) + \mathrm{nullity}(\mathbf{A}) = n$.
5. **Unique solution:** consistent system has a unique solution $\iff$ $\mathrm{rank}(\mathbf{A}) = n$.
6. **Surjectivity:** $\mathbf{Ax} = \mathbf{b}$ consistent for all $\mathbf{b} \in \mathbb{R}^m$ $\iff$ $\mathrm{rank}(\mathbf{A}) = m$.
7. **Invertibility (square):** $\mathbf{A}$ invertible $\iff$ $\mathrm{rank}(\mathbf{A}) = n$ $\iff$ $\mathrm{nullity}(\mathbf{A}) = 0$.